

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	DNA: plasmid, closed circular, CsCl purified x2; Co-transfection with 2 plasmids; pHIV LTR-CAT and pSV40-Tat; also transfection with pCH110 alone (7.2 kB).
Species Used	Human, U937, hystiocytic lymphoma		

Before the Pulse

Cell Growth Medium	RPMI 1640 + 10% fetal calf serum + penicillin/streptomycin, L-glutamine, sodium pyruvate (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Log growth
		Pre-pulse Incubation	10 min at 4° C.

Wash Solution Cell growth media

The Pulse

Instruments Used Gene Pulser® apparatus & Capacitance

Electroporation Temperature	4 °C	Cuvette Gap	0.4 cm
Electroporation Medium*	Cell growth media	Voltage	0.30 kV
Cell Density	1.3 x 10 ⁷ / ml	Field Strength	0.75 kV/cm
Volume of Cells	250 µl	Capacitor	960 µF
DNA Concentration	Not given	Resistor	(Pulse Controller) Ω none
DNA Resuspension Buffer	Not given	Time Constant	Not given
Volume of DNA	Not given		

After the Pulse

Outgrowth Medium Cell growth media

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	24 to 48 hr.
Selection Method or Assay Used	CAT assay, stain for β-galactosidase
Electroporation Efficiency	About 500 transfectants / µg DNA
Per Cent Survival	15%

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